nnUNet model performance comparison

(Advanced Data Analysis & Artificial Intelligence)

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Research Question

How do different training strategies affect the performance of nnUNet?

01

Dataset Size

How does image size for training influence model performance ?

02

Cross-Validation

Single fold vs. full 5-fold cross-validation

03

Augmentation

Synthetic data expansion as an alternative to real images

Dataset & Preparation

Image Acquisition

- ✓ Photos of various types of pens
- ✓ Ground-truth masks created manually via Roboflow
- ✓ 2 classes: 0 = Background, 1 = Pen
- ✓ PNG format for nnUNet compatibility
- ✓ 9 test images (completely new, unseen pens)

84
Base Training Images

9
Test Images (unseen pens)

2
Classes (Background / Pen)

Annotation Tool: Roboflow
Masks exported as PNG

Model Overview

Model	Training Images	Fold(s)	Strategy	Note
M1	84	Fold 0	Baseline	Base experiment
M2	42	Fold 0	Fewer Data	Random subset of 84
M3	21	Fold 0	Very Few Data	Random subset of 42
M4	84	Fold 0–4	5-Fold CV	Full cross-validation
M5	168	Fold 0	Augmentation	2× via Albumentations

All models trained for 100 epochs (hard stop — loss curves converged well within this limit)

Evaluation & Metrics

Test set: 9 completely new, unseen pen images — ground-truth masks created manually via Roboflow

Dice Score

$$2 \cdot |A \cap B| / (|A| + |B|)$$

Measures overlap between predicted and ground-truth mask. Range 0–1, higher is better.

IoU (Jaccard Index)

$$|A \cap B| / |A \cup B|$$

Ratio of intersection to union. Stricter than Dice, also 0–1.

Per-Image Dice Score

Model	Img 1	Img 2	Img 3	Img 4	Img 5	Img 6	Img 7	Img 8	Img 9	Mean
M1 84 imgs · Fold 0	0.516	0.580	0.286	0.502	0.653	0.327	0.864	0.584	0.511	**0.5107**
M2 42 imgs · Fold 0	0.422	0.437	0.136	0.375	0.555	0.293	0.854	0.301	0.450	**0.3828**
M3 21 imgs · Fold 0	0.153	0.267	0.150	0.290	0.309	0.194	0.449	0.260	0.396	**0.2572**
M4 84 imgs · 5-Fold CV	0.565	0.571	0.305	0.499	0.735	0.351	0.880	0.501	0.531	**0.5237**
M5 168 imgs · Fold 0	0.548	0.647	0.338	0.430	0.726	0.415	0.865	0.648	0.432	**0.5377**

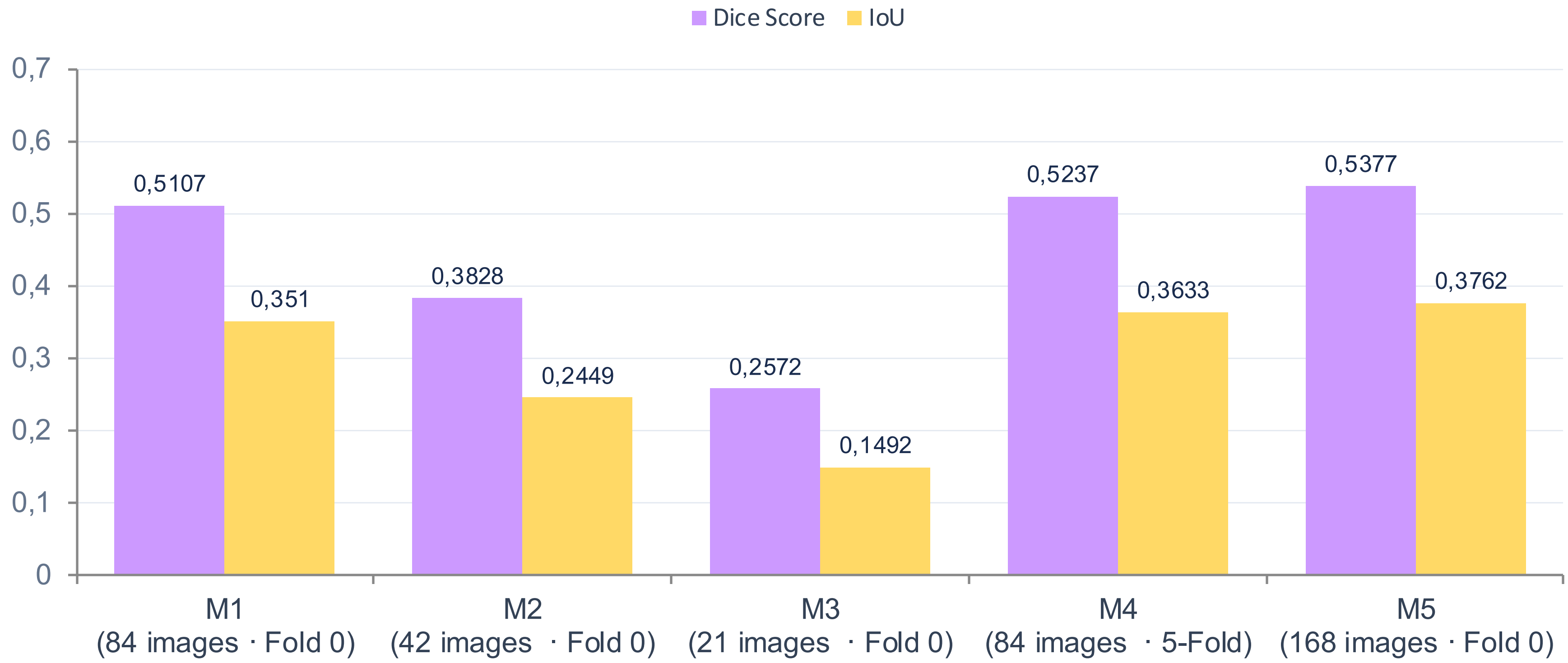
lower score *higher score*

Per-Image IoU Score

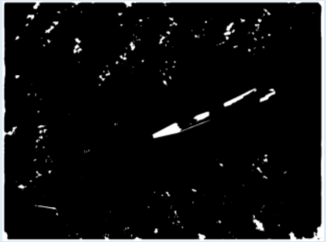
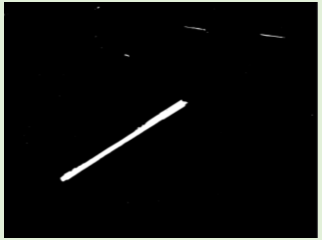
Model	Img 1	Img 2	Img 3	Img 4	Img 5	Img 6	Img 7	Img 8	Img 9	Mean
M1 84 imgs · Fold 0	0.348	0.409	0.167	0.335	0.485	0.195	0.760	0.412	0.343	**0.3510**
M2 42 imgs · Fold 0	0.268	0.280	0.073	0.231	0.384	0.172	0.746	0.177	0.290	**0.2449**
M3 21 imgs · Fold 0	0.083	0.154	0.081	0.170	0.183	0.107	0.289	0.149	0.247	**0.1492**
M4 84 imgs · 5-Fold CV	0.393	0.400	0.180	0.332	0.580	0.213	0.786	0.335	0.361	**0.3633**
M5 168 imgs · Fold 0	0.377	0.478	0.203	0.274	0.570	0.262	0.763	0.480	0.276	**0.3762**

lower score — *higher score*

Results — Mean Performance per Model



Results – visual comparison

	Img 3	Img 5	Img 7
original image			
M3 prediction *(worst performing model)*			
M5 prediction *(best performing model)*			
	(Image all models performed worst on)		*(Image all models performed best on)*

Thank you for your Attention!